Introduction

Having learned about propositions and how to combine them, we now turn our attention to quantifiers and implications. Quantifiers are how we describe mathematical quantities, while implications are how we describe mathematical consequences.

Learning Objectives

- Define the two types of quantifiers, describe their differences, and demonstrate how to prove quantified statements.
- Describe the difference between multiply quantified statements when the order of the quantifiers is changed.
- Define what is meant by an implication, its converse, and its contrapositive.
- Identify the equivalence of an implication and its contrapositive to prove implications.
- Correctly negate sentences with quantifiers and implications.

Quantifiers

Quantifiers allow us to discuss the number of objects which satisfy a predicate. If we wish to discuss *every* element of a set, we use the *universal quantifier* $\forall$, read as "for all." To state that an element in a set *exists*, we use the *existential quantifier* $\exists$, read as "there exists."

When combined with a predicate P, we can assign truth values to quantified statements. For example, let S be a universe of discourse. The statement $\forall x \in S, P(x)$ will be true precisely when $P(x)$ is true for every element in S. On the other hand, $\exists x \in S, P(x)$ will be true as long as one (or more) element of S makes $P(x)$ true.[1]

The addition of quantifiers to a predicate creates a proposition, such as the following:

- Every cow has a favourite radio station.
- There is a black horse.
- In every sport, there exists someone who breaks the rules.
- There is one textbook that every class uses.
- There is no largest real number.

These last two examples have multiple quantifiers. Can you spot them?

[1] A simple mnemonic for remembering which symbol corresponds to which quantifier is that "for all" looks like an upside down A which stands for ALL, and "there exists" looks like a backwards E which stands for EXISTS.

Exercise 1 Write the above sentences by creating predicates and using appropriate quantifiers.

Exercise 2 Consider the proposition "$\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, y^2 = x$." Is this sentence true? Describe this proposition in English without using any variable names.

Example 3

Determine whether each of the quantified statements is true or false.

1. $\forall x \in \mathbb{Z}^*, x^2 \geq 0$
2. $\exists x \in \mathbb{R}, x = \sqrt{-1}$,
3. $\forall x \in \mathbb{Q}, \forall y \in \mathbb{Q}, x + y \in \mathbb{Q}$,
4. $\exists x \in \mathbb{Z}^+, \exists y \in \mathbb{Z}^+, x/y \in \mathbb{Z}^+$.

Solution.

1. This statement is true, since squaring any non-zero number results in a positive number.
2. This statement is false. If such an x existed, it would also satisfy $x^2 = -1$. By our comment in part 1, the square of a non-zero number is always positive, leading us to a contradiction.
3. This statement is true. Suppose that x and y are arbitrary rational numbers, and write these as $x = a/b$ and $y = c/d$ for appropriate choices of $a, c \in \mathbb{Z}$ and $b, d \in \mathbb{Z}^+$. Adding these together we get
$$x + y = \frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}.$$
Since $ad + bc \in \mathbb{Z}$ and $bd \in \mathbb{Z}^+$, this is also a rational number.
4. This statement is true. For example, by setting $x = 4$ and $y = 2$ we have $x/y = 4/2 = 2$, which is also a positive integer. ■

Notice how the above solutions demonstrated the truth of quantifier statements. To show that $\exists x \in S, P(x)$, we find a single example of an $x \in S$ which makes $P(x)$ true. To show that $\forall x \in S, P(x)$ is more subtle. Rather than try to demonstrate $P(x)$ for every x, we choose an *arbitrary* $x \in S$. If $P(x)$ is true for an arbitrary x, then it must be true for every x.

Doubly quantified statements must be treated with caution. One may freely interchange two adjacent quantifiers of the *same* type, but not of different type. For example, the statements

$$\forall x \in \mathbb{Q}, \forall y \in \mathbb{Q}, x+y \in \mathbb{Q} \quad \text{is logically equivalent to} \quad \forall y \in \mathbb{Q}, \forall x \in \mathbb{Q}, x+y \in \mathbb{Q},$$

and

$$\exists x \in \mathbb{Z}^+, \exists y \in \mathbb{Z}^+, x/y \in \mathbb{Z}^+ \quad \text{is logically equivalent to} \quad \exists y \in \mathbb{Z}^+, \exists x \in \mathbb{Z}^+, x/y \in \mathbb{Z}^+.$$

However, interchanging existential and universal quantifiers can lead to serious trouble.

Example 4

Consider the statements

$$\forall x \in \mathbb{R}, \exists y \in \mathbb{R},\ x + y = 0 \tag{1}$$

and

$$\exists x \in \mathbb{R}, \forall y \in \mathbb{R},\ x + y = 0. \tag{2}$$

Compare these expressions by translating them as follows:

1. Convert the mathematical notation into English.
2. Turn the sentence derived above into a simple sentence, which does not involve any variables.
3. Evaluate whether each statement is true or false.

Solution. We start with equation (1) for which a direct translation of the notation into English gives us

"For all x in the real numbers, there exists y in the real numbers, (such that) $x + y = 0$."

This is fine but not very enlightening. By recognizing that $x + y = 0$ is equivalent to $x = -y$, we could also re-interpret this sentence as saying "For every real number there is another real number which is its negative." Dropping the superfluous words we arrive at the intuitive statement

"Every real number has a negative."

This statement is certainly true: Given a real number a, we can construct its negative to be $-a$.

Looking at (2) we have

$$\exists y \in \mathbb{R}, \forall x \in \mathbb{R},\ x + y = 0. \tag{3}$$

Using the same translation process as above, the corresponding simple sentence is given by

"There is an element which is the negative of every real number."

This says there is a number to which we can add any other number and always get zero. Certainly this is not true! If it were, then there would be a number n such that $n + a = 0$ and $n + b = 0$ for any real numbers a and b. Equating these expressions, we would find that $n + a = n + b$ which in turn implies that $a = b$. This would force all real numbers to be equal, which is nonsense. ■

Example 4 teaches us that changing the order of the quantifiers significantly changes the logical statement, and hence the truth of that statement. To borrow a term from the computer scientists, universal quantifiers admit a 'scope' to the existential quantifiers they precede. For example, the statement $\forall x, \exists y, P(x, y)$ means that the choice of y is allowed to depend upon x. The statement $\exists y, \forall x, P(x, y)$ does not confer this dependence: the choice of y must work for every x.[2]

> **Exercise 5** Let S be the set of all students in a classroom, and $B(a, b)$ be the statement "student a has the same birthday as student b." Write the mathematical statements
>
> $$\forall a \in S, \forall b \in S, B(a, b), \quad \forall a \in S, \exists b \in S, B(a, b)$$
>
> $$\exists a \in S, \forall b \in S, B(a, b), \quad \exists a \in S, \exists b \in S, B(a, b)$$
>
> in plain language, avoiding the use of variables if possible.

> **Exercise 6** Find a predicate P that we have not already used, such that one of the following is true and one is false:
>
> $$\forall x, \exists y, P(x, y) \quad \text{and} \quad \exists x, \forall y, P(x, y).$$

Note that while we've used the symbols "$\forall, \exists$" a lot to emphasize their differences, you would normally avoid symbols when stating mathematical results. For example, a theorem might be stated as "For every positive real number x, there is a real number y such that $x = y^2$." Moreover, students traditionally struggle with statements that involve multiple quantifiers. It is good practice to translate such sentences into "plain English" and to work with many examples.

Negating Quantifiers

To develop intuition for negating quantifiers, let's think about how we would disprove a statement involving a quantifier. For example, the universally quantified statement "every horse is black" may be disproved by showing that there exists a non-black horse. Mathematically, if $P(x)$ is "x is a black horse,"

$$\text{the negation of} \quad \forall x, P(x) \quad \text{is} \quad \exists x, \neg P(x).$$

The existentially quantified statement "there exists a pink horse" is disproved by showing that "every horse is not pink." Mathematically, if $P(x)$ is the statement "x is a pink horse," then

$$\text{the negation of} \quad \exists x, P(x) \quad \text{is} \quad \forall x, \neg P(x).$$

[2] Certainly it doesn't get any worse than two quantifiers, right? You probably learned what it means for a function f to be (everywhere) continuous in high school. When you get to the heart of that definition, it involves no less than five quantifiers, all of which alternate between being universal and existential quantifiers. See Exercise 18 for an example.

By thinking about the case of a general predicate P, the negation rules above still apply.

Example 7

Consider the mathematical statement $\forall x \in \mathbb{R}, x < x^2$. Determine whether this sentence is true or false, and write the negation of this sentence.

Solution. This sentence is false. For example, if $x = 1/2$ then $x^2 = 1/4$, showing that $x > x^2$. The negation of this sentence is

$$\exists x \in \mathbb{R} : x \geq x^2.$$

Notice that our counter-example satisfies the negation of our sentence, as we would expect. ■

Exercise 8 Write the following propositions using quantifiers, and then negate the sentence:

1. There is a real number whose square is negative.
2. Every real number has a cubed root.
3. There is a smallest positive real number.

Implications

At the core of mathematical statements are implications, which consist of 'if-then' statements. A typical theorem contains a *hypothesis* and a *conclusion*, such that IF the hypothesis is TRUE, THEN the conclusion is TRUE. This is a conditional statement that requires us to first check the truth of the hypothesis before we can ascertain the truth of the conclusion, and is called an implication because the veracity of the first statement implies the veracity of the second.

To frame this mathematically, let P and Q be predicates. The statement "If P then Q," or alternatively "P implies Q," is written $P \implies Q$ and has truth table given by Table 1.

IMPLICATION

P	Q	$P \implies Q$
T	T	T
T	F	F
F	T	T
F	F	T

Table 1: The truth table for the implication $P \implies Q$.

Carefully consider the bottom two rows of Table 1, which are known as *vacuous truths*. The idea is that a universally false hypothesis P can have any implication it wants, since that implication will never be tested. For example, consider the statement

"If pigs can fly, then the sky is black."

This is a true statement because whenever pigs can fly then the sky is black. This may seem artificial and contrived, but vacuous truths appear in mathematics frequently, so it is important to be aware of how they are handled.

Example 9

Let $D(x)$ be the predicate "x is a dog" and let $A(x)$ be the predicate "x is an animal." Consider the truth of the implications $D(x) \implies A(x)$, $A(x) \implies D(x)$ and $\neg A(x) \implies \neg D(x)$.

Solution. The arguments are given below:

$D(x) \implies A(x)$	This is the statement "If x is a dog then it is an animal" or "All dogs are animals" and is TRUE.
$A(x) \implies D(x)$	This is the statement "If x is an animal then it is a dog" or "All animals are dogs." A cat is an animal which is not a dog, so this implication must be FALSE.
$\neg A(x) \implies \neg D(x)$	This is the statement "If x is not an animal then it is not a dog," and is a TRUE sentence. Indeed, if x is not an animal then it could not be a dog. If it were a dog, then by our first implication, it would be an animal and this would be a contradiction. ■

Definition 10

Let P and Q be predicates with the implication $P \implies Q$.

- The *contrapositive* of $P \implies Q$ is the statement $\neg Q \implies \neg P$.
- The *converse* of $P \implies Q$ is the statement $Q \implies P$.

Example 9 shows that the converse of true statement is not necessarily true. As for the contrapositive, we have the following result:

Proposition 11

If P and Q are predicates, then $P \implies Q$ and $\neg Q \implies \neg P$ are logically equivalent.

Exercise 12 Show that Proposition 11 is true by constructing the corresponding truth table for $\neg Q \implies \neg P$.

Because $P \implies Q$ and $\neg Q \implies \neg P$ are logically equivalent, that means we can use the contrapositive to prove theorems!

Proposition 13

Let n be an integer. If n^2 is even, then n is even.

Proof. Given the predicates

$$P(n) : n^2 \text{ is even,} \quad \text{and} \quad Q(n) : n \text{ is even,}$$

we want to show that $P(n) \implies Q(n)$. By Proposition 11, it suffices to show $\neg Q(n) \implies \neg P(n)$; or equivalently, "If n is not even, then n^2 is not even." Since being odd is the same thing as being not even, this boils down to showing

"If n is odd, then n^2 is odd."

Take a moment and compare this to our original statement, and think about the fact that these are effectively the same statement. Suppose then that n is odd, and write $n = 2k + 1$ for some integer $k \in \mathbb{Z}$. Now

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1 = 2\ell + 1,$$

where $\ell = 2k^2 + 2k$ is an integer. Thus n^2 is odd, as required. $\square$

When both $P \implies Q$ and its converse $Q \implies P$ are true, we write $P \iff Q$ and say that "P is true *if and only if* Q is true." This means that the statements P and Q are logically equivalent: whatever truth value $P(x)$ has, $Q(x)$ will have the same. It is difficult at this point to give examples of "if and only if" statements that are not just trivial restatements of one another, but some examples might include:

- An integer $n \in \mathbb{Z}$ is even if and only if $n/2$ is an integer.
- An integer $n \in \mathbb{Z}$ is divisible by 10 if and only if its one's digit is a 0
- A triangle is isosceles if and only if at least two of its angles are equal.

The words 'necessary' and 'sufficient' are often used to indicate the direction of an implication. If P and Q are predicates, then

- "P is a necessary condition for Q" is the implication $Q \implies P$,
- "P is a sufficient condition for Q" is the implication $P \implies Q$,
- "P is necessary and sufficient for Q" is the statement $P \iff Q$.

Exercise 14 Let $n \in \mathbb{Z}^*$. Show that n is even if and only if n^2 is even. Conclude that n is odd if and only if n^2 is odd.

Negating an Implication

In Example 9 we found that the statement $A(x) \implies D(x)$, read as "Every animal is a dog," was false. To show that it was false we used the example of a cat, which is an animal but is not a dog. More generally, if P and Q are predicates in a universe S, we say that $x \in S$ is a *counter-example* to $P \implies Q$ if $P(x)$ is true but $Q(x)$ is not true; that is, $P(x) \wedge \neg Q(x)$ is true. Counter-examples are exactly how implications are negated.

Proposition 15

If P and Q are predicates, the negation of the implication $P \implies Q$ is the statement $P \wedge \neg Q$.

Exercise 16 Prove Proposition 15 by constructing the truth tables for $\neg(P \implies Q)$ and $P \wedge \neg Q$ and showing they are the same.

Example 17

Negate the following sentence: "If x is a duck, then x likes peanut butter."

Solution. Here we have the predicates $P(x) =$"x is a duck" and $Q(x) =$"x likes peanut butter" with the sentence above being the implication $P \implies Q$. The negation of this implication is $P \wedge \neg Q$, or "x is a duck and x does not like peanut butter." ■

Exercise 18 In calculus, we say that a function f is (everywhere) continuous if

$$\forall c \in \mathbb{R}, \exists L \in \mathbb{R}, \forall \epsilon > 0, \exists \delta > 0, \forall x \in \mathbb{R}, |x - c| < \delta \implies |f(x) - L| < \epsilon.$$

Negate this sentence to determine the mathematical statement that a function is not continuous. (We won't be proving this sort of thing in this class. You can take MAT137 or MAT157 if you're interested in doing that.)

A Note on Language

Having now covered the basics of logic, it's important to point out a few places where mathematical language disagrees with everyday English.

OR: Mathematics uses *inclusive-or*, while everyday language usually uses *exclusive-or*. To demonstrate this, suppose someone said to you "I'm going to eat either Pizza or Thai food for dinner." Implicit in this statement is that the person will eat one of these things, not both of them. But mathematically, this statement would include the case where the person orders them both.

NOT: In English, you can generally negate a sentence by putting the word "not" wherever you like. For example, "Not everything is alright" and "Everything is not alright" are both the same in English. Mathematically, these statements are different. If $A(x)$ is the statement "x is alright" then these sentences are

$$\neg\left(\forall x, A(x)\right) = \exists x, \neg A(x) \quad \text{and} \quad \forall x, \neg A(x), \quad \text{respectively.}$$

These have completely different meanings! One says "Something is not okay" and the other says "Nothing is okay." If that example feels artificial to you, we can turn to Shakespeare:

"All that glisters is not gold" – Merchant of Venice, Act II Scene 7.

"Glisters" here is often changed to "glistens," and the sentence means "Not everything that shines is valuable." Despite being a well-know phrase, in English we'd be more likely to say "Not everything that glisters is gold." Note the different placements of the word "not," and how these statements have different mathematical interpretations.

IMPLICATION: Most If-Then sentences in English are secretly if-and-only-if statements. For example, if your mother said to you "If you do the dishes, then you can go out tonight," there's an implicit understanding that if you don't do the dishes, you don't get to go out. This implicit statement is the converse of what your mother said! If you're having trouble seeing this, let D be the proposition "you do the dishes" and O be the proposition "you go out tonight." Your mother said to you "$D \implies O$" but you also understand that implicit in her statement is the fact that "$\neg D \implies \neg O$", which is logically equivalent to "$O \implies D$." Hence your mother is saying "$D \iff O$," even though it would sound weird if she said to you "You will do the dishes if and only if you go out tonight."

Moreover, implications in English are often *causal*, in the sense that they describe a relationship between the hypothesis and conclusion. I might say "If it's raining today, then I'm not going outside," and we understand that the reason I'm not going outside is because it's raining. Most of the mathematical statements we see in the course will also be causal in the same sense, but strictly speaking they don't have to be. It is perfectly reasonable in mathematics to say "If it's raining today, then an asteroid will hit Jupiter." This sentence could be true, even though there isn't an obvious link between the hypothesis and conclusion.

All of this is to remind you to be careful when translating between natural language and mathematics, since they generally don't agree. Remember, mathematics is precise. When in doubt, return to the definition.

Exercises

1. For each of the following logical statements, produce a plain English translation. Your translation should avoid the use of variable names and negation

as much as possible. For example, the statement $\forall x \in \mathbb{Z}^*, x^2 \geq 0$ might translate to "The square of any non-negative number is positive."

(a) $\exists x \in \mathbb{Z}^+, \exists y \in \mathbb{Z}^+, x^2 - 2y^2 = 3$
(b) $\forall x \in \mathbb{Z}, \exists y, z \in \mathbb{Z}, x = yz$
(c) $\forall x \in \mathbb{R}, \forall y \in \mathbb{R}, xy < 0 \implies (x < 0 \lor y < 0)$

2. Let P be the set of polynomials. Determine whether each of the following statements is true or false. If the statement is true, prove it. If the statement is false, provide a counter-example.

(a) $\forall x \in \mathbb{R}, \forall p \in P, p(x) = 0$
(b) $\forall x \in \mathbb{R}, \exists p \in P, p(x) = 0$
(c) $\exists x \in \mathbb{R}, \forall p \in P, p(x) = 0$
(d) $\exists p \in P, \forall x \in \mathbb{R}, p(x) = 0$
(e) $\forall p \in P, \exists x \in \mathbb{R}, p(x) = 0$
(f) $\exists x \in \mathbb{R}, \exists p \in P, p(x) = 0$

3. Create the truth tables for each of the following logical statements:

(a) $P \iff Q$
(b) $P \implies (Q \lor \neg P)$
(c) $P \implies (Q \iff \neg R)$

4. Are the statements $P \implies (Q \implies R)$ and $(P \implies Q) \implies R$, logically equivalent?

5. Negate each of the following logical statements (implication and quantifier)

(a) $\forall x, [P(x) \implies Q(x)]$
(b) $\exists x, [P(x) \iff Q(x)]$
(c) $\exists x, [(P(x) \land Q(x)) \implies R(x)]$
(d) $\forall x, [P(x) \implies (Q(x) \implies R(x))]$

6. Determine whether the following statements are logically equivalent:

- $[\forall x, P(x)] \implies [\forall x, Q(x)]$
- $\forall x, [P(x) \implies Q(x)]$

7. The following is a correct statement that we will learn later in the course. Let $\mathbb{P}$ denote the set of prime numbers.

$$\forall p \in \mathbb{Z}^+, \big[p \in \mathbb{P} \iff \big(\forall a \in \mathbb{Z}, \forall b \in \mathbb{Z}, (\exists k \in \mathbb{Z}, pk = ab) \implies \exists \ell \in \mathbb{Z}\,[p\ell = a \lor p\ell = b]\,\big)\big]$$

Write the negation of this sentence.

8. Write the contrapositive and converse of each (not necessarily true) statement below:

(a) If $x^2 + y^2 = 4$ then $-2 \leq x \leq 2$.
(b) If the product of two real numbers is rational, then both of its factors must be rational.
(c) An integer which is divisible by 2 cannot be prime.
(d) I will pass this course if I receive a good mark on Term Test 1.